The Impact of Recreational Marijuana Legalization on Crime Rates

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Abstract

This study investigates the impact of recreational marijuana legalization on violent and property crime rates across U.S. states from 2010 to 2022 using a fixed effect model. Surprisingly, the analysis reveals unexpected outcomes in some areas. While the anticipated effect of legalization on violent crime rates lacked statistical significance, an inverse correlation between education levels and violent crime rates emerged as a significant finding, aligning with prevailing sociological and economic theories. Intriguingly, the impact of poverty rates on crime rates did not exhibit expected trends, warranting further exploration. However, the legalization's influence on property crime rates, specifically theft and robbery, was notably significant, challenging certain theoretical presumptions.

These findings hold implications for policy formulation, societal understanding, and potential business strategies. They emphasize the nuanced nature of crime causation and the importance of considering multifaceted factors in policy-making. Moreover, the inverse correlation between education and crime rates underscores the potential role of educational initiatives in fostering societal stability. However, the unexpected results regarding poverty rates signify the complexity of crime dynamics, necessitating more comprehensive investigations. Overall, this study offers critical insights into the intricate relationship between marijuana legalization and crime rates, prompting further exploration into multifaceted influences on criminal behavior.

Introduction

The project aims to examine the effect of marijuana legalization on crime rates, with a particular focus on comparing crime rate data before and after the legalization of recreational marijuana in a specific state, such as California or Massachusetts.

The crime rate data for this project will be obtained from Uniform Crime Statistics. Crime Statistics can provide a state-level-panel dataset with a longitudinal component. Since states where marijuana is legalized do not legalize it at the same time, the project will analyze crime rates in all U. S. states from 2010 to 2022. Each dataset will observe the crime rates in a state for several years before and after marijuana legalization.

The aim of this project is that The implementation of marijuana legalization policies is expected to lead to a decrease in specific types of crimes, particularly those related to the illegal sale and possession of marijuana. This will significantly reduce the size of the illicit market. As a result of policy reform, law enforcement resources will be reallocated, leading to enhanced policing activities in other areas and thus a reduction in certain types of crime rates. However, marijuana legalization also brings

forth certain social issues. Underage marijuana abuse can impair mental and physical health. Adult marijuana abuse can lead to decreased judgment and behavior that affects public safety. Misuse of marijuana can increase certain types of illegal activities. Investigating the impact of legalization will help policymakers in states where marijuana remains illegal to assess the effectiveness of such policies and determine if they are worth considering. This research can also assist states where marijuana is legal in evaluating whether there is a need to strengthen regulations governing the purchase of marijuana to enhance societal well-being.

Review of the literature

In 2012, with the passage of Amendment 64 and Initiative 502, Colorado and Washington became the first states in the United States to legalize recreational Marijuana. Since then, the legalization of recreational Marijuana has been expanding across the United States. Whether or not legalization will have a potential and actual impact on the public health of the community, especially on crime rates, has become the focus of public attention on this major policy change. Studies by the National Institutes of Health have clearly shown that 30% of Marijuana users experience smoking dependence and withdrawal symptoms from Marijuana, which are more pronounced in adolescents (National Insitiute of Health, 2020). The abuse of Marijuana has also caused much grief for parents whose children have lost their lives to Marijuana at a young age (Stack, 2021). Some citizens believe that the legalization of Marijuana as a gateway drug has caused users, including minors, to become more addicted, thus exacerbating crime (Science world, 2019). However, numerous studies have found that Marijuana has a low potential for addiction and a relatively low impact on public health compared to other drugs and is not a gateway drug among teens (Hall & Degenhardt, 2009; Budeny et al, 2007). Proponents of Marijuana legalization argue that the implementation of legalization policies will reduce certain types of crime, particularly those related to the illegal sale and possession of Marijuana (Dragone et al., 2020). They also argue that legalization will reallocate law enforcement resources to increase policing efforts in areas other than drug incidents, thereby reducing the rate of certain types of crime (Trilling, 2016).

There is an ongoing debate about the impact of recreational Marijuana legalization on crime rates, and various studies that have examined this issue have yielded mixed results. As the first states to legalize recreational Marijuana, Colorado and Washington have seen positive developments in changes in crime rates (Harper & Jorgensen, 2022). Evidence suggests that legalization has a optimistic crime-reducing effect (Wu et al., 2020; Makin et al., 2018). There is even less need to worry about social safety around retail dispensaries, as data support that the addition of a dispensary to a neighborhood can result in a reduction of 17 crimes per 10,000 residents per month, which equates to an average crime reduction of about 19% over the sample period (Brinkman & Lamme, 2019; Burkhardt & Goemans, 2018). Dr. Freisthler et al. found that the density of Marijuana outlets in a local or neighborhood area was not associated with violent crime rates by collecting data from 481 crime

locations and Marijuana outlets over a 34-month period (Freisthler, 2017). In addition to Colorado and Washington, states such as California and New York have also seen a decrease in crime rates as a result of legalization (Hunt, 2018; Chu & Townsend, 2019; Lu et al., 2018).While most studies point to Marijuana policy reform as beneficial to public health and safety, not all states that have legalized have seen such optimistic results. Various cities in Oregon have experienced significant increases in a variety of serious crimes compared to non-legalized states (Wu et al, 2021). Additionally, Marijuana abuse occurs at significantly higher rates among adolescents (Kerr et al., 2017).

The fact that the results of the Oregon crime rate study are inconsistent with the results of the previous studies for Colorado and Washington suggests that more specific research is needed to address this issue. What's more, this also reflects the limitations of these studies in that data from different states (or different regions) need to be analyzed in a targeted manner, and conclusions about the effects of legalization cannot be drawn solely from the results of data from individual states. It is worth noting that policy reform to legalize Marijuana does not happen overnight. A state must go through the process of legalizing medical Marijuana, legalizing Marijuana possession, legalizing Marijuana production, and legalizing Marijuana sales to fully implement a recreational Marijuana legalization policy. Each step affects the change in crime rates to a greater or lesser extent, so studies must use the full data interval regardless of the area being studied. This will allow for more accurate results.

In sum, as opposed to the rest of the literature, which only analyzes crime rates for individual specific states, this project favors a more complete mix of data, i.e., the collection of crime rates (which includes both violent and property crimes) for all U.S. states from two years prior to the time of the first U.S. state's legalization of marijuana (2010) to the year 2022.

Theoretical Model

The impact of recreational marijuana legalization on crime rates has been the focus of many studies. Looking at this issue from multiple theoretical perspectives can provide a richer and more comprehensive explanation. First, according to deterrence theory, marijuana legalization may lead to a decrease in crime rates. Legalization reduces the illicit demand for marijuana, decreasing the opportunities for related illegal transactions and criminal activities.

However, another theory holds the opposite view that legalizing marijuana may lead to an increase in other non-violent crimes. According to the alternative theory, legalization frees up police and judicial resources, which may allow the police to focus more on other crimes and therefore may lead to an increase in other types of crime. This may be due to the fact that legalization of marijuana spreads out the fight against other crime types to a certain extent, resulting in a shift in the distribution of crime.

Furthermore, from the perspective of social and economic factors, the legalization of marijuana may have a complex impact on the social structure and crime rates. It

may have altered employment opportunities and the economic structure, providing new economic opportunities for some, thereby reducing some of the drivers of nonviolent crime. However, legalization may also have some negative effects, such as social instability or other economic and social challenges, which may in turn have an impact on crime rates.

These theories provide different perspectives and expectations of the impact that marijuana legalization may have on crime rates. However, these theories are also subject to a certain amount of controversy and uncertainty; therefore, empirical research on these theories will allow us to better understand the actual impacts of marijuana legalization on crime rates and whether these impacts are consistent with theoretical expectations.

Data and Variables

For this project, a comprehensive all-state-panel dataset spanning the years 2010-2022 underwent analysis, encompassing a total of 663 observations (the dataset begins in 2010 since it is 2 years before the first states legalized marijuana for recreational use in 2012; the reason why the dataset has 663 observations is that according to the state FIPS code, there are 51 "states" in the U.S., and the dataset contents 13 years - 13 * 51 = 663 observations). Given the enactment of recreational marijuana legalization at the state level, the primary unit of analysis is the state.

The key dependent variables of this project are various crime rates, specifically violent crimes encompassing total violent crime, homicide, rape, robbery, and aggravated assault, as well as property crime involving total property crime, burglary, larceny theft, and motor vehicle theft. However, due to considerable skewness and kurtosis exhibited by these variables, natural logarithms were applied for normalization. For instance, the initial skewness and kurtosis of homicide were 3.44 and 24.53 respectively, which decreased to 0.04 and 3.10 after applying natural logarithms.

The primary independent variable under scrutiny in this poject is the legislation pertaining to recreational marijuana use, as indicated by a dummy variable for each state/year observation. Notably, considering the time lag for dispensaries to commence the sale of recreational marijuana subsequent to its legalization, the true treatment year is considered as the year following the legislative year. For instance, if Massachusetts sanctioned the legal purchase of recreational marijuana products in retail stores in 2016, the corresponding dummy variable for Massachusetts would be 0 from 2010 to 2018, and would transition to 1 from 2019 to 2022.

It is anticipated that numerous factors may influence the crime rate in a state. Given that data is collected at the state level, the model will also control for characteristics specific to that state, including the average income, unemployment rate, education level, poverty rate, population, and law enforcement presence. Data were compiled from multiple sources, including FBI and state government websites, US Census Bureau tables, ST.LOUIS FED, U.S. Bureau of Labor Statistics, and comprehensive internet searches.

Empirical Methodology

The empirical methodology of this project involves the use of panel data analysis to examine the impact of recreational marijuana legalization on crime rates in US states. This approach allows for a comprehensive assessment through longitudinal aspects to observe changes in crime rates before and after marijuana legalization legislation.

The estimated equation takes the following form:ln_crime it = β 0 + β 1(treat_recre_legal) it + β 2(ln_population) it + β 3(poverty) it + β 4(edu_level) it + β 5(ln_avg_incm) it + β 6(unemp) it + β 7(ln_law_enforce) it + Ui + Rt + E. Where Ui represents the panel_variable - state_FIPS_code of this panel dataset, Rt represents this time variable - year. enforce) it + Ui + Rt + E. where Ui represents the panel_variable - state_FIPS_code for this panel dataset, and Rt represents the time variable - year.

Of these variables, I convert the population, average income, police force, and y variables to natural log.The natural logarithm is used to convert these variables to address skewness and kurtosis to ensure that the estimates are more robust and reliable. The sign of the coefficients is theoretically uncertain. While some theories suggest that crime rates may decline after legalization due to a decrease in black market activity or a shift in law enforcement priorities. Based on deterrence theory, which suggests that marijuana legalization may lead to a decrease in crime rates, I predict that independent variables in which a recreational marijuana legalized state has a smaller population, a higher poverty rate, a higher level of education, a higher average income, a lower unemployment rate, and a higher police force, the state's crime rate is likely to be lower (where "average income "and population" are somewhat analogous). Other theories, however, suggest that crime rates may rise due to lenient attitudes toward drug use or changes in criminal behavior.

Given the nonlinear nature of the relationship between marijuana legalization and crime rates, nonlinear forms, such as interaction terms between legalization and other contextual factors, could be explored. Estimation techniques would involve panel data regressions, such as fixed effects or random effects models, to account for unobserved heterogeneity across states and years.

Results

If the independent variable is ln_violent_crime, the following is the result of the prediction:

```
. xtreg ln_violent_crime treat_recre_legal ln_population poverty edu_level ln_avg_incm unemp
ln_law_enforce i.year, fe i(state_FIPS_code)
    Fixed-effects (within) regression              Number of obs     =        663
    Group variable: state_FIPS~e                   Number of groups  =         51
    R-squared:                                     Obs per group:
```

```
     Within  = 0.1176                                      min =        13
     Between = 0.0241                                      avg =      13.0
     Overall = 0.0241                                      max =        13
                                                F(19, 593)        =      4.16
corr(u_i, Xb) = -0.7841                         Prob > F          =    0.0000
------------------------------------------------------------------------------
 ln_violent_crime | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
treat_recre_legal |   .0111636   .0193975     0.58   0.565    -.0269326    .0492597
    ln_population |   .5036011   .2237888     2.25   0.025      .064086    .9431162
          poverty |  -.0017325   .0036696    -0.47   0.637    -.0089394    .0054745
        edu_level |  -.0357843    .007055    -5.07   0.000    -.0496401   -.0219285
      ln_avg_incm |   .0832917   .1105633     0.75   0.452    -.1338515     .300435
            unemp |   .0216043   .0054908     3.93   0.000     .0108206     .032388
   ln_law_enforce |   .0046542   .0160425     0.29   0.772     -.026853    .0361613
                  |
             year |
             2011 |  -.0055603   .0222306    -0.25   0.803    -.0492205    .0380999
             2012 |   .0351257   .0241436     1.45   0.146    -.0122918    .0825431
             2013 |   .0519843   .0260713     1.99   0.047      .000781    .1031876
             2014 |   .0597408   .0302167     1.98   0.048      .000396    .1190855
             2015 |   .1345674   .0337621     3.99   0.000     .0682597    .2008752
             2016 |   .1919831   .0376231     5.10   0.000     .1180925    .2658737
             2017 |   .2140218   .0422538     5.07   0.000     .1310364    .2970071
             2018 |   .2053972   .0465076     4.42   0.000     .1140575    .2967369
             2019 |    .214586   .0520153     4.13   0.000     .1124294    .3167425
             2020 |   .2276615   .0540462     4.21   0.000     .1215162    .3338069
             2021 |   .2437738   .0584492     4.17   0.000     .1289811    .3585664
             2022 |   .2614866   .0666862     3.92   0.000     .1305168    .3924564
                  |
            _cons |  -1.869845   3.408183    -0.55   0.583    -8.563422    4.823732
------------------+-----------------------------------------------------------
          sigma_u |   .6642362
          sigma_e |  .11060035
              rho |  .97302316   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(50, 593) = 117.87                   Prob > F = 0.0000
```

For the violent-crime model, there are several areas of concern that may be surprising or worth exploring in depth:

1. treat_recre_legal (treatment for recreational legalization) has a coefficient of 0.011, a standard error of 0.019, a t-value of 0.58, and a p-value of 0.565. Despite the positive coefficient, the relatively small t-value and high p-value suggest that it may not be statistically significant. This may be surprising, as this

variable was expected to have a significant effect on violent crime rates.

2. edu_level (level of education) has a coefficient of -0.036, a t-value of -5.07, and a p-value of 0.000, indicating that it is statistically significant and of the expected sign. This is an important finding which implies that there may be a negative correlation between education level and violent crime rates.

3. poverty (poverty rate) has a coefficient of -0.002, a t-value of -0.47, and a p-value of 0.637, the coefficient of this variable is not significant, which may be contrary to expectations. This result does not seem to give clear evidence on the relationship between poverty rate and violent crime rate.

4. The non-significance of police strength and average income is also surprising.

If the independent variable is ln_property_crime, the following is the result of the prediction:

```
. xtreg ln_property_crime treat_recre_legal ln_population poverty edu_level ln_avg_incm unemp
ln_law_enforce i.year, fe i(state_FIPS_code)
Fixed-effects (within) regression               Number of obs     =        663
Group variable: state_FIPS~e                    Number of groups  =         51
R-squared:                                      Obs per group:
     Within  = 0.7592                                         min =         13
     Between = 0.0030                                         avg =       13.0
     Overall = 0.0177                                         max =         13
                                                F(19, 593)        =      98.40
corr(u_i, Xb) = -0.9516                         Prob > F          =     0.0000
------------------------------------------------------------------------------
ln_property_crime | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
treat_recre_legal |   .0763872   .0168722     4.53   0.000     .0432506    .1095239
    ln_population |   .8725648   .1946548     4.48   0.000     .4902681    1.254862
          poverty |  -.0011787   .0031919    -0.37   0.712    -.0074474      .00509
        edu_level |   -.028265   .0061365    -4.61   0.000     -.040317    -.016213
      ln_avg_incm |   .0150163   .0961696     0.16   0.876    -.1738581    .2038907
            unemp |   .0289993    .004776     6.07   0.000     .0196194    .0383791
   ln_law_enforce |  -.0050845    .013954    -0.36   0.716    -.0324899    .0223209
                  |
             year |
            2011  |   .0152244   .0193365     0.79   0.431    -.0227519    .0532006
            2012  |   .0480965   .0210005     2.29   0.022     .0068522    .0893409
            2013  |   .0206781   .0226772     0.91   0.362    -.0238593    .0652155
            2014  |  -.0040051   .0262829    -0.15   0.879     -.055624    .0476139
            2015  |  -.0127415   .0293668    -0.43   0.665     -.070417     .044934
            2016  |  -.0017226   .0327251    -0.05   0.958    -.0659938    .0625485
            2017  |  -.0170058    .036753    -0.46   0.644    -.0891877     .055176
```

```
        2018 |  -.0680852   .040453   -1.68   0.093   -.1475338   .0113635
        2019 |  -.1062942  .0452436   -2.35   0.019   -.1951515  -.0174369
        2020 |  -.2667161  .0470102   -5.67   0.000   -.3590429  -.1743893
        2021 |  -.2549563    .05084   -5.01   0.000   -.3548046  -.1551079
        2022 |  -.1681954  .0580046   -2.90   0.004   -.2821149  -.0542759
             |
       _cons |   -4.80579  2.964488   -1.62   0.106   -10.62796   1.016383
------------------+----------------------------------------------------------------
     sigma_u |  .95751984
     sigma_e |  .09620182
         rho |  .99000669   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(50, 593) = 62.55                    Prob > F = 0.0000
```

There are several aspects of the regression results that may be surprising or worthy of further exploration:

1. It is worth noting that the coefficient on treat_recre_legal is 0.076, significant and positive. This suggests that property crime rates appear to have increased following the legalization of marijuana. This is contrary to the deterrence theory hypothesis, as some believe that legalization may reduce property crime rates by reducing black market activity and changing police enforcement priorities.
2. The coefficient on the variable education_level is negative, significant and in the expected direction, suggesting that higher levels of education may be associated with lower crime rates. This is consistent with the general view in sociology and economics that higher levels of education lead to higher levels of social stability and the rule of law, which reduces crime.
3. The effect of poverty level on property crime rates is not significant. This statement seems contradictory.
4. The signs of the coefficients on population size (ln_population) and unemployment (unemp) are consistent with expectations, but there may also be some complex interactions or factors that are not taken into account.

In order to complete the project more rigorously, I tried to compare the fixed effect model with the random effect model to find a better way to simulate this project:

```
. quietly xtreg ln_violent_crime treat_recre_legal ln_population poverty edu_level ln_avg_incm
unemp ln_law_enforce i.year, fe i(state_FIPS_code)
. est store fe
. quietly xtreg ln_violent_crime treat_recre_legal ln_population poverty edu_level ln_avg_incm
unemp ln_law_enforce i.year, re theta
. est store re
. hausman fe re, sigmamore
```

```
Note: the rank of the differenced variance matrix (7) does not equal the number of coefficients being tested (19); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

                    ---- Coefficients ----
             |      (b)          (B)            (b-B)     sqrt(diag(V_b-V_B))
             |       fe           re         Difference       Std. err.
-------------+----------------------------------------------------------------
treat_recr~l |    .0111636     .0079091        .0032545        .0027874
ln_populat~n |    .5036011     .0501486        .4534525        .2270215
     poverty |   -.0017325     .0004279       -.0021604        .0006341
   edu_level |   -.0357843    -.0178746       -.0179097        .0050168
 ln_avg_incm |    .0832917     .1188568       -.0355651        .0256009
       unemp |    .0216043     .0233331       -.0017288        .0009439
ln_law_enf~e |    .0046542     .0126994       -.0080453        .0023371
        year |
       2011  |   -.0055603    -.0094831        .0039228         .001908
       2012  |    .0351257     .0249686        .0101571        .0044715
       2013  |    .0519843     .0385296        .0134547        .0064127
       2014  |    .0597408     .0434166        .0163242        .0081901
       2015  |    .1345674      .113725        .0208425        .0107807
       2016  |    .1919831     .1626016        .0293815        .0141572
       2017  |    .2140218     .1743661        .0396557        .0176656
       2018  |    .2053972     .1586636        .0467336        .0206469
       2019  |     .214586     .1609156        .0536704         .024218
       2020  |    .2276615     .1374663        .0901952        .0331685
       2021  |    .2437738     .1552739        .0884999          .03385
       2022  |    .2614866     .1646043        .0968823        .0367954
------------------------------------------------------------------------------
                           b = Consistent under H0 and Ha; obtained from xtreg.
            B = Inconsistent under Ha, efficient under H0; obtained from xtreg.
Test of H0: Difference in coefficients not systematic
    chi2(7) = (b-B)'[(V_b-V_B)^(-1)](b-B)
            =  49.57
Prob > chi2 = 0.0000
```

Since the Prob > chi2 = 0.0000, the random effect model's assumption that Ui is totally random is rejected. Therefore, using fixed model is better than the other one.

Property crimes include burglary, larceny, and motor vehicle theft. by replacing ln_burglary, ln_larceny, and ln_motor_vehicle_theft with the y variable of the fixed effect model, respectively, it can be concluded that the recreational marijuana legalization has a significant effect on burglary and larceny crime rates in the United States. As shown in the following data.

```
. xtreg ln_burglary treat_recre_legal ln_population poverty edu_level ln_avg_incm unemp
ln_law_enforce i.year, fe i(state_FIPS_code)
Fixed-effects (within) regression               Number of obs     =        663
Group variable: state_FIPS~e                    Number of groups  =         51
R-squared:                                      Obs per group:
     Within  = 0.8573                                         min =         13
     Between = 0.1115                                         avg =       13.0
     Overall = 0.1355                                         max =         13
                                                F(19, 593)        =     187.51
corr(u_i, Xb) = -0.9524                         Prob > F          =     0.0000
------------------------------------------------------------------------------
      ln_burglary | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
treat_recre_legal |   .0759947   .0255517     2.97   0.003     .0258118    .1261776
    ln_population |   1.371882   .2947897     4.65   0.000     .7929231    1.950841
          poverty |   .0034188   .0048338     0.71   0.480    -.0060747    .0129123
        edu_level |  -.0857321   .0092933    -9.23   0.000    -.1039839   -.0674803
      ln_avg_incm |   .1548425   .1456414     1.06   0.288    -.1311933    .4408783
            unemp |   .0504941   .0072328     6.98   0.000      .036289    .0646992
   ln_law_enforce |  -.0063175   .0211323    -0.30   0.765    -.0478208    .0351858
                  |
             year |
            2011  |    .050825   .0292836     1.74   0.083    -.0066872    .1083371
            2012  |    .108968   .0318036     3.43   0.001     .0465065    .1714294
            2013  |   .0750412   .0343429     2.19   0.029     .0075928    .1424897
            2014  |   .0189656   .0398034     0.48   0.634    -.0592073    .0971384
            2015  |   .0235626   .0444737     0.53   0.596    -.0637825    .1109077
            2016  |    .037252   .0495596     0.75   0.453    -.0600818    .1345857
            2017  |   .0145249   .0556596     0.26   0.794     -.094789    .1238387
            2018  |  -.0655442    .061263    -1.07   0.285     -.185863    .0547746
            2019  |  -.1385764    .068518    -2.02   0.044    -.2731439   -.0040089
            2020  |  -.3024851   .0711934    -4.25   0.000    -.4423069   -.1626633
            2021  |  -.3172369   .0769933    -4.12   0.000    -.4684495   -.1660243
            2022  |  -.2522349   .0878435    -2.87   0.004    -.4247571   -.0797126
                  |
            _cons |  -14.00945   4.489488    -3.12   0.002    -22.82668   -5.192215
------------------+-----------------------------------------------------------
          sigma_u |  1.4779559
          sigma_e |  .14569025
              rho |  .99037638   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(50, 593) = 37.95                    Prob > F = 0.0000
.
. xtreg ln_larceny treat_recre_legal ln_population poverty edu_level ln_avg_incm unemp
```

```
ln_law_enforce i.year, fe i(state_FIPS_code)
Fixed-effects (within) regression               Number of obs     =        663
Group variable: state_FIPS~e                    Number of groups  =         51
R-squared:                                      Obs per group:
     Within  = 0.7618                                         min =         13
     Between = 0.0014                                         avg =       13.0
     Overall = 0.0055                                         max =         13
                                                F(19, 593)        =      99.83
corr(u_i, Xb) = -0.9338                         Prob > F          =     0.0000
------------------------------------------------------------------------------
       ln_larceny | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
treat_recre_legal |    .073624   .0151221     4.87   0.000     .0439247    .1033234
    ln_population |   .6766622   .1744632     3.88   0.000     .3340213    1.019303
          poverty |  -.0025338   .0028608    -0.89   0.376    -.0081523    .0030846
        edu_level |   -.013298      .0055    -2.42   0.016    -.0240998   -.0024962
      ln_avg_incm |  -.0465743   .0861939    -0.54   0.589    -.2158567    .1227081
            unemp |   .0183978   .0042805     4.30   0.000     .0099909    .0268047
   ln_law_enforce |  -.0037982   .0125066    -0.30   0.761    -.0283608    .0207643
                  |
             year |
            2011  |   .0057125   .0173307     0.33   0.742    -.0283245    .0397495
            2012  |   .0320175   .0188221     1.70   0.089    -.0049486    .0689835
            2013  |   .0064966   .0203249     0.32   0.749    -.0334209    .0464141
            2014  |  -.0167373   .0235566    -0.71   0.478    -.0630018    .0295271
            2015  |  -.0365129   .0263205    -1.39   0.166    -.0882057    .0151799
            2016  |  -.0391244   .0293305    -1.33   0.183    -.0967287    .0184799
            2017  |  -.0624209   .0329406    -1.89   0.059    -.1271153    .0022735
            2018  |  -.1141587   .0362568    -3.15   0.002    -.1853661   -.0429513
            2019  |  -.1436852   .0405505    -3.54   0.000    -.2233252   -.0640451
            2020  |   -.306643   .0421338    -7.28   0.000    -.3893927   -.2238934
            2021  |  -.3223853   .0455663    -7.08   0.000    -.4118764   -.2328943
            2022  |   -.248874   .0519878    -4.79   0.000    -.3509765   -.1467715
                  |
            _cons |  -1.856386    2.65698    -0.70   0.485    -7.074622    3.361849
------------------+-----------------------------------------------------------
          sigma_u |  .76019406
          sigma_e |  .08622276
              rho |  .98729884   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(50, 593) = 70.41                    Prob > F = 0.0000
.
. xtreg ln_motor_vehicle_theft treat_recre_legal ln_population poverty edu_level ln_avg_incm
unemp ln_law_enforce i.year, fe i(state_FIPS_code)
```

```
Fixed-effects (within) regression               Number of obs     =        663
Group variable: state_FIPS~e                    Number of groups  =         51
R-squared:                                      Obs per group:
     Within  = 0.2263                                         min =         13
     Between = 0.0224                                         avg =       13.0
     Overall = 0.0234                                         max =         13
                                                F(19, 593)        =       9.13
corr(u_i, Xb) = -0.9029                         Prob > F          =     0.0000
------------------------------------------------------------------------------
ln_motor_vehicl~t | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
------------------+-----------------------------------------------------------
treat_recre_legal |   .0318984   .0311528     1.02   0.306    -.0292848    .0930816
    ln_population |   .9800089   .3594088     2.73   0.007     .2741399    1.685878
          poverty |   .0008447   .0058934     0.14   0.886    -.0107298    .0124192
        edu_level |  -.0646936   .0113304    -5.71   0.000    -.0869463    -.042441
      ln_avg_incm |  -.0544452   .1775666    -0.31   0.759    -.4031812    .2942908
            unemp |   .0452589   .0088183     5.13   0.000       .02794    .0625778
   ln_law_enforce |   -.027034   .0257646    -1.05   0.294     -.077635     .023567
                  |
             year |
            2011  |   .0189243   .0357027     0.53   0.596    -.0511947    .0890434
            2012  |    .072741   .0387751     1.88   0.061    -.0034123    .1488942
            2013  |   .0796338    .041871     1.90   0.058    -.0025996    .1618672
            2014  |   .1364451   .0485285     2.81   0.005     .0411364    .2317538
            2015  |   .2166354   .0542225     4.00   0.000     .1101439    .3231269
            2016  |     .35789   .0604233     5.92   0.000     .2392204    .4765597
            2017  |   .4509111   .0678604     6.64   0.000     .3176352     .584187
            2018  |   .4820027   .0746921     6.45   0.000     .3353095    .6286959
            2019  |   .4790155   .0835374     5.73   0.000     .3149503    .6430808
            2020  |   .4826783   .0867992     5.56   0.000      .312207    .6531496
            2021  |   .6530312   .0938705     6.96   0.000     .4686722    .8373903
            2022  |   .8152388   .1070992     7.61   0.000      .604899    1.025579
                  |
            _cons |  -7.298606   5.473602    -1.33   0.183    -18.04861    3.451398
------------------+-----------------------------------------------------------
          sigma_u |  1.1467395
          sigma_e |  .17762614
              rho |   .9765692   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(50, 593) = 74.82                    Prob > F = 0.0000
```

Conclusion

By using a FIXED EFFECT MODEL to simulate this effect of legalizing

recreational marijuana on violent and property crime rates in all states across the U.S. from 2010 to 2022, we conclude:

1. The results of the effect of recreational marijuana legalization on violent crime rates are surprising. Despite the positive coefficient, the small t-value and high p-value suggest that it may not be statistically significant, which is inconsistent with the expected significant effect of this variable on violent crime rates.
2. The coefficient on the possible negative correlation between education level and violent crime rate is significant and in the expected direction. This finding suggests that an increase in the level of education may reduce the rate of violent crime, which is consistent with the general view in sociology and economics.
3. The coefficient of the effect of poverty rate on the violent crime rate is not significant, which may not be consistent with expectations.
4. The results for property crime rates show that the legalization of recreational marijuana has a significant effect on burglary and robbery crime rates. Although this contradicts some theoretical assumptions, it suggests the possibility that legalization may have led to a number of social and economic changes that affected property crime rates.

Not only do these findings overturn some previous analyses of crime rates in states that have only legalized recreational marijuana in a very few states, but they also guide policymakers to smarter legislation. In addition, recognizing the impact of education on crime rates may emphasize the importance of educational initiatives for social stability and safety. However, the unexpected results on poverty rates and the lack of significance in some areas highlight the complexity of crime causation and urge further research to gain comprehensive insights.

Table 1

Statistics

	Variable	Description	Mean	Standard Deviation	Min	Max	Sample Size
Key Independent Variables	**treat_recre_legal**	Dummy variable that equals 1 if the state is recreational marijuana legal and starting from the following year	0.134238	0.341165	0	1	663
	ln_population	Natural log of state population Population measured in person	6318964	7130928	564554	3.95E+07	
	poverty	State poverty level measured in %	12.59864	3.558889	3.7	25.8	
	edu_level	State education level which people with bachelor or higher education level Education level measured in %	31.27391	6.708508	17.5	65.4	
	ln_avg_incm	Natural log of state average annual income measured in $	60643.86	12885.12	35520	108200	
	unemp	State average annual unemployment rate measured in %	5.773002	2.262423	2.1	13.7	
	ln_law_enforce	Natural log of state number of law enforcement officers Law enforce measured in person	18591.77	22551.82	9	121251	
Key Dependent Variable	**ln_violent_crime**	natural log of violent crime rate crime rate is measured in per 100000 people	5.847595	0.426867	4.63074	7.190526	
	ln_homicide	natural log of homicide crime rate is measured in per 100000 people	1.463144	0.600331	-0.1278	3.71284	
	ln_rape_revised	natural log of rape which is revised crime rate is measured in per 100000 people	3.667041	0.385493	2.41143	5.105157	
	ln_robbery	natural log of robbery crime rate is measured in per 100000 people	4.137635	0.752886	2.06812	6.572255	
	ln_aggravated_assault	natural log of aggravated assault crime rate is measured in per	5.413851	0.479216	4.06663	6.472053	

		100000 people					
	ln_property_crime	natural log of property crime rate crime rate is measured in per 100000 people	7.752653	0.308135	6.83179	8.551608	
	ln_burglary	natural log of burglay crime rate is measured in per 100000 people	6.031942	0.502766	4.29674	7.054078	
	ln_larceny	natural log of larceny crime rate is measured in per 100000 people	7.424299	0.282366	6.51780	8.312986	
	ln_motor_vehicle_theft	natural log of motor vehicle theft crime rate is measured in per 100000 people	5.284863	0.522785	3.38506	6.728222	

Table 2

Violent Crime & Property Crime Mentions - Baseline Model

Dependent Variables	**Natural log of violent crime**	**Natural log of property crime**
treat_recre_legal	.0111636 (0.0193975)	0.0763872*** (0.0168722)
ln_population	0.5036011** (0.0411)	0.8725648*** (0.1946548)
poverty	-0.0017325 (0.0036696)	-0.0011787 (0.0031919)
edu_level	-0.035784*** (0.007055)	-0.028265*** (0.0061365)
ln_avg_incm	-0.832917 (0.1105633)	-0.0150163 (0.0961696)
unemp	-0.021604*** (0.0054908)	0.0289993*** (0.004776)
ln_law_enforce	0.0046542 (0.0160425)	-0.0050845 (0.013954)
Year Effects	Yes***	Yes***
Crime Effects	Yes***	Yes***
Crime-Linear Trend	Yes	Yes
R-Squared	0.0241	0.0177
Observations	663	663

Notes: Dependent variables are natural log of violent crime and property crime. Standard errors are adjusted and are reported in parentheses. *** significant at 1 percent, ** significant at 5 percent, and * significant at 10 percent. Elasticities are reported where the own-price coefficient is significant at 10 percent or less in a one-tailed test.

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